

ZEEROCODES

DSS Software Requirements Specification

Production system specification

DIGITAL SALES SCHOOL · PRODUCTION SPECIFICATION

1. ARCHITECTURE

Recommended: React/Next.js + TypeScript frontend, application/domain services, PostgreSQL, object storage, background jobs, payment gateway, notification service and an AI gateway abstraction.

Presentation → Application → Domain → Persistence, with AI, payments, notifications and analytics behind explicit interfaces.

2. FUNCTIONAL REQUIREMENTS

FR-001	Authentication and RBAC
FR-002	Admissions/application
FR-003	Payment and server-side enrollment
FR-004	Curriculum/module state
FR-005	Locked-module authorization
FR-006	Lesson progress
FR-007	Quiz engine
FR-008	Assessment engine
FR-009	Assignment submission/review
FR-010	Prerequisite unlock engine
FR-011	AI role-play
FR-012	AI tutor
FR-013	Readiness Index
FR-014	Portfolio
FR-015	Certification
FR-016	Internship/deployment
FR-017	Analytics
FR-018	Audit logging

3. LOCK/UNLOCK STATE MACHINE

LOCKED → prerequisite incomplete → LOCKED LOCKED → prerequisite passed → UNLOCKED
UNLOCKED → started → IN_PROGRESS IN_PROGRESS → all completion rules passed →
COMPLETED COMPLETED → next module becomes UNLOCKED

4. QUIZ ENGINE

- Configurable question types.
- Versioned questions.
- Attempt history.
- Configurable pass threshold; default 80%.
- Retry/cooldown.
- Server-side score calculation.
- No client-side completion authority.

5. ASSESSMENT ENGINE

- Rubric-based scoring.
- Pass/fail or numeric modes.
- AI may provide provisional grading.
- Human assessor can confirm/override where configured.
- Overrides require reason, actor and timestamp.

6. AI ROLE-PLAY

AI buyers should include CEO, CTO, CFO, marketing director, HR director, procurement manager, sceptical founder and price-sensitive customer. The curriculum requires scoring questioning, listening, product knowledge, discovery, value articulation, objections, closing and professionalism. ■filecite■turn950file3■L1-L12■

- Persist every turn.
- Use approved scenario bank.
- Use hidden buyer objective/context.
- Return rubric feedback.
- Allow replay.
- Never issue authoritative certification status.

7. SECURITY

- Server-side resource authorization.
- Prevent IDOR.
- Rate-limit auth/quiz/AI endpoints.
- Protect recordings/submissions.
- Encrypt sensitive data.
- Audit privileged actions.
- Keep AI system prompts server-side.

8. API SURFACE

POST /api/applications; POST /api/payments/checkout; POST /api/payments/webhook; GET /api/student/dashboard; GET /api/curriculum; GET /api/modules/:id; POST /api/modules/:id/lessons/:lessonId/complete; POST

/api/quizzes/:id/attempts; POST /api/assignments/:id/submissions; POST /api/roleplay/sessions; POST /api/roleplay/sessions/:id/messages; GET /api/readiness; POST /api/portfolio/artifacts; GET /api/certification/status; POST /api/admin/completion-overrides.

9. TESTING

- Unit tests for unlock rules/scoring.
- Integration tests for payment→enrollment.
- Security tests for locked-module bypass.
- E2E student journey.
- AI rubric evaluation tests.
- Regression tests for admin overrides and certification.